



Jarms Construction

SERVICE



TAT : RESOURCE MANAGEMENT

EXCAVATION AND LINING OPERATIONS

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Table of Content

1	Introduction	7	PERT Analysis
2	Scope and stakeholders	8	Findings
3	Work Breakdown Structure	9	Probability of finishing 15% early and crashing by 10%
4	Project Activity Breakdown	10	Monte Carlo Analysis
5	Gantt Chart & Critical path	11	Conclusion
6	Network Diagram	12	Q&A

Project Introduction and Objectives

- The TAT : Resource Management is a sub-project of the “Trans under Atlantic Tunnel” focusing on the supply and management of construction resources required in building a section of the tunnel to ensure safety, functionality, and efficiency.
- In this project, we are taking inspiration from the Agile methodology of project management where we would break down the construction process of the bigger tunnel into various iterations and implement an incremental approach of operations towards each iteration.
- The current presentation encompasses the plan and estimations for one iteration.
- Emphasising structural integrity, cost-effectiveness, quality compliance, risk mitigation and timely completion this project is aimed to align with best practices in the industry.



Stakeholders & Project Scope

- Excavation
- Lining and sealing
- Design & Planning
- Quality Assurance
- Maintenance & Control



Internal Stakeholders

- Project Managers
- Engineers (Civil, Mechanical, Electrical)

Skilled Labourers

- Carry out tasks related to excavation, lining, and equipment management.

External Stakeholders

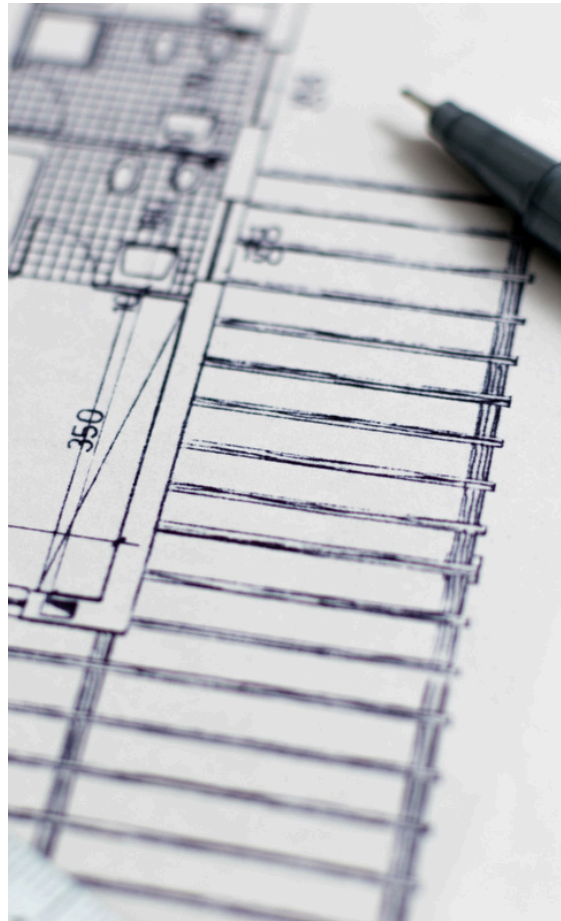
- Government Authorities

Contractors and Suppliers

- Community and Environmental



Work breakdown structure



DESIGN & PLANNING



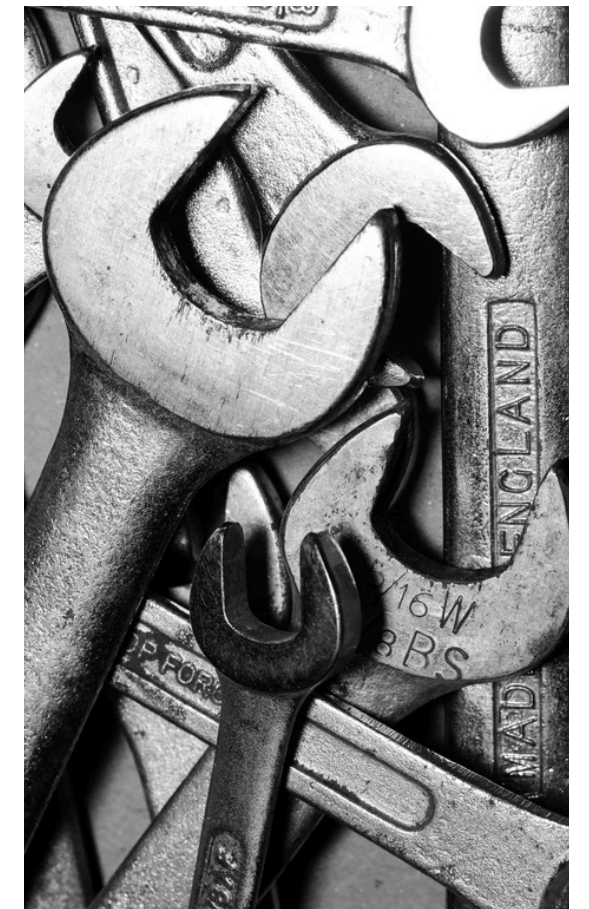
EXCAVATION



LINING AND SEALING



QUALITY ASSURANCE



MAINTENANCE & CONTROL



Accessories

Finishing Touches: Must-Have Accessories for the Season

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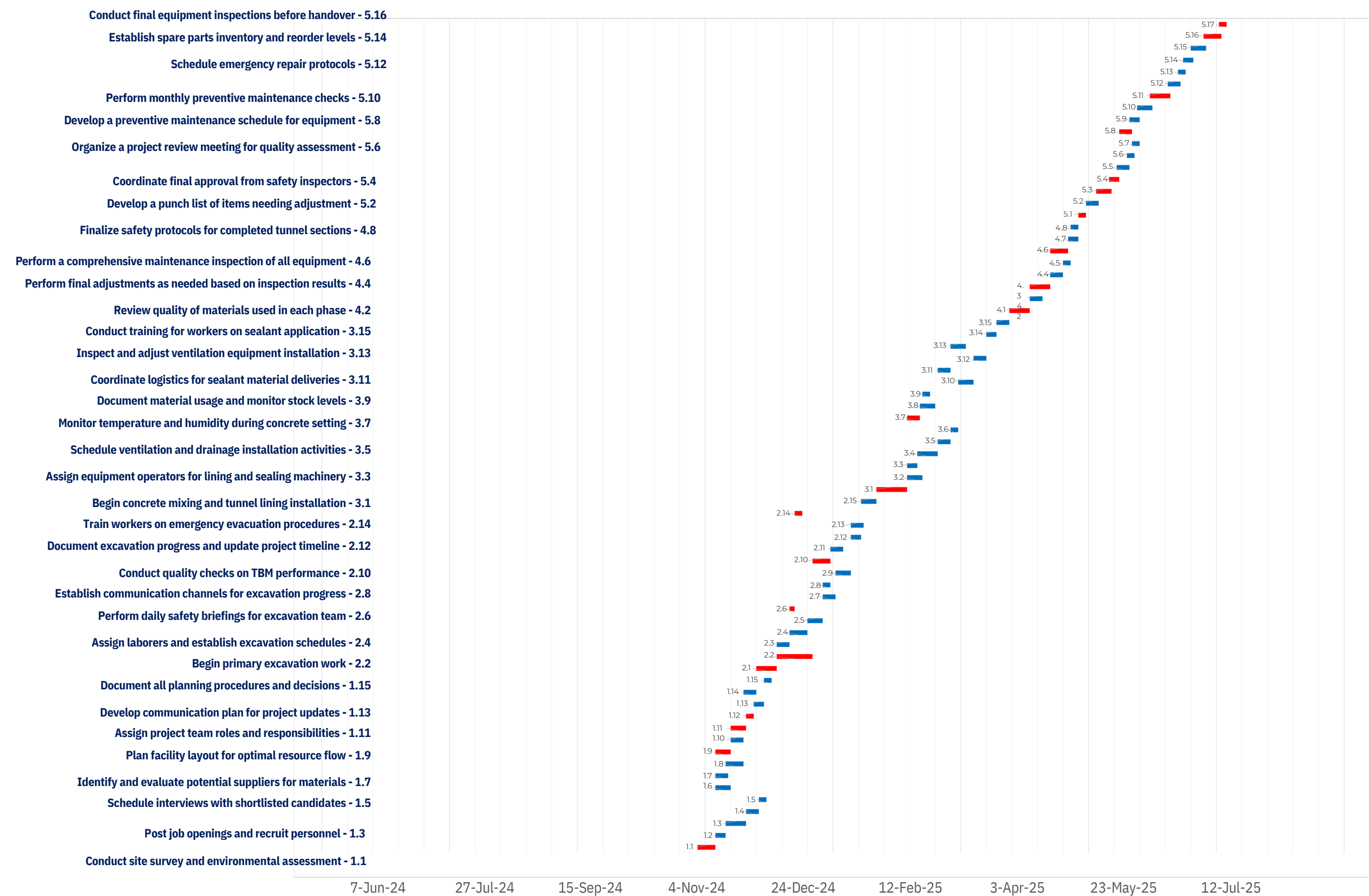
1.1 Conduct site survey and environmental assessment	2.1 Set up and test the Tunnel Boring Machine (TBM)	3.1 Begin concrete mixing and tunnel lining installation	4.1 Conduct structural and safety inspections for completed sections	5.1 Update equipment records and document condition for handover
1.2 Define required skill sets for each role	2.2 Begin primary excavation work	3.2 Monitor inventory levels of concrete and sealing materials	4.2 Review quality of materials used in each phase	5.2 Develop a punch list of items needing adjustment
1.3 Post job openings and recruit personnel	2.3 Implement safety protocols for excavation	3.3 Assign equipment operators for lining and sealing machinery	4.3 Conduct system testing for tunnel safety	5.3 Conduct follow-up inspections to ensure all adjustments are complete
1.4 Screen applicants for qualifications	2.4 Assign labourers and establish excavation schedules	3.4 Establish waterproofing procedures and apply sealants	4.4 Perform final adjustments as needed based on inspection results	5.4 Coordinate final approval from safety inspectors
1.5 Schedule interviews with shortlisted candidates	2.5 Arrange for material deliveries and debris removal logistics	3.5 Schedule ventilation and drainage installation activities	4.5 Document results of quality inspections and system tests	5.5 Perform mock drills for emergency preparednessl
1.7 Identify and evaluate potential suppliers for materials	2.6 Perform daily safety briefings for excavation team	3.6 Conduct safety checks for lining and sealing operations	4.6 Perform a comprehensive maintenance inspection of all equipment	5.6 Organize a project review meeting for quality assessment
1.6 Develop safety protocols and initial risk assessment	2.7 Monitor material usage and reorder levels for excavation	3.7 Monitor temperature and humidity during concrete setting	4.7 Schedule additional repairs or adjustments as needed	5.7 Finalise and distribute quality inspection reports
1.8 Negotiate and finalize contracts with suppliers	2.8 Establish communication channels for excavation progress	3.8 Perform bi-weekly maintenance on lining equipment	4.8 Finalize safety protocols for completed tunnel sections	5.8 Develop a preventive maintenance schedule for equipment
1.9 Plan facility layout for optimal resource flow	2.9 Coordinate waste removal and recycling during excavation	3.9 Document material usage and monitor stock levels		5.9 Assign maintenance personnel for routine tasks
1.10 Define equipment specifications and requirements	2.10 Conduct quality checks on TBM performance	3.10 Review and inspect installed lining for quality control		5.10 Perform monthly preventive maintenance checks
1.11 Assign project team roles and responsibilities	2.11 Perform weekly equipment maintenance and repairs	3.11 Coordinate logistics for sealant material deliveries		5.11 Conduct quarterly inspections of heavy machinery
1.12 Conduct project kickoff meeting	2.12 Document excavation progress and update project timeline	3.12 Update project timeline based on sealing phase progress		5.12 Review and update equipment maintenance logs
1.13 Develop communication plan for project updates	2.13 Coordinate transportation for debris disposal	3.13 Inspect and adjust ventilation equipment installation		5.13 Establish spare parts inventory and reorder levels
1.14 Schedule initial maintenance setup for key equipment	2.14 Train workers on emergency evacuation procedures	3.14 Establish fire safety protocols for sealing materials		5.14 Train maintenance staff on critical equipment handling
1.15 Document all planning procedures and decisions	2.15 Adjust excavation plan based on ground conditions	3.15 Conduct training for workers on sealant application		5.15 Conduct final equipment inspections before handover
				5.16 Archive all maintenance and project documentation

Project Activity Breakdown

ID	Activity	Predecessor(s)	O	M	P	E	Variance	Critical Path
1.1	Conduct site survey and environment al assessment	-	5	7	10	7.33	0.69	Yes
1.2	Define required skill sets for each	1.1	3	4	6	4.00	0.25	No
1.3	Post job openings and recruit	1.2	6	8	12	8.33	1.00	No
1.4	Screen applicants	1.3	4	5	7	5.00	0.11	No
1.5	Schedule interviews with shortlisted	1.4	2	3	5	3.00	0.25	No
1.6	Develop safety protocols and initial risk	1.1	4	6	9	6.17	0.69	No
1.7	Identify and evaluate potential suppliers for	1.1	3	5	7	5.00	0.44	No
1.8	Negotiate and finalize contracts	1.7	5	7	10	7.33	0.69	No
1.9	Plan facility layout for optimal	1.1	4	6	8	6.00	0.44	Yes
1.10	Define equipment specification	1.9	3	5	7	5.00	0.44	No
1.11	Assign project team roles and	1.6, 1.9	4	6	8	6.00	0.44	Yes
1.12	Conduct project	1.11	2	3	4	3.00	0.11	Yes
1.13	Develop communicati on plan for project	1.12	3	4	6	4.00	0.25	No
1.14	Schedule initial maintenance setup for key	1.10	4	5	7	5.00	0.11	No
1.15	Document all planning procedures and decisions	1.13	2	3	5	3.00	0.25	No
2.1	Set up and test the Tunnel	1.14	5	8	12	8.33	1.00	Yes
2.2	Begin primary	2.1	10	14	20	14.67	2.78	Yes
2.3	Implement safety protocols for	2.1	3	5	8	5.17	0.69	No
2.4	Assign laborers and establish excavation	2.3	5	7	10	7.33	0.69	No
2.5	Arrange for material deliveries and debris removal	2.4	4	6	9	6.17	0.69	No
2.6	Perform daily safety briefings for excavation	2.3	1	2	3	2.00	0.11	Yes
2.7	Monitor material usage and reorder	2.5	3	5	7	5.00	0.44	No
2.8	Establish communicati on channels for excavation	2.5	2	3	5	3.00	0.25	No
2.9	Coordinate waste removal and recycling	2.7, 2.8	4	6	9	6.17	0.69	No
2.10	Conduct quality checks on	2.2	5	7	10	7.33	0.69	Yes
2.11	Perform weekly equipment maintenance	2.10	3	5	8	5.17	0.69	No
2.12	Document excavation progress and update project	2.9, 2.11	2	4	6	4.00	0.44	No
2.13	Coordinate transportatio n for debris	2.9	3	5	8	5.17	0.69	No
2.14	Train workers on emergency evacuation	2.6	2	3	4	3.00	0.11	Yes
2.15	Adjust excavation plan based	2.12	4	6	9	6.17	0.69	No
3.1	Begin concrete mixing and	2.15	8	12	16	12.00	1.78	Yes
3.2	Monitor inventory levels of concrete anda	3.1	4	6	9	6.17	0.69	No
3.3	Assign equipment operators for lining and	3.1	2	4	6	4.00	0.44	No
3.4	Establish waterproofin g procedures and apply	3.3	6	8	12	8.33	1.00	No
3.5	Schedule ventilation and drainage installation activities	3.4	3	5	8	5.17	0.69	No

ID	Activity	Predecessor(s)	O	M	P	E	Variance	Critical Path
3.6	Conduct safety checks for lining and sealing	3.2, 3.5	2	3	5	3.00	0.25	No
3.7	Monitor temperature and humidity during	3.1	3	5	7	5.00	0.44	Yes
3.8	Perform bi- weekly maintenance on lining	3.7	4	6	8	6.00	0.44	No
3.9	Document material usage and monitor	3.2	2	3	4	3.00	0.11	No
3.10	Review and inspect installed lining for	3.6, 3.8	4	6	9	6.17	0.69	No
3.11	Coordinate logistics for sealant material	3.4	3	5	7	5.00	0.44	No
3.12	Update project timeline based on	3.10	3	5	8	5.17	0.69	No
3.13	Inspect and adjust ventilation equipment	3.5	4	6	9	6.17	0.69	No
3.14	Establish fire safety protocols for sealing	3.12	2	4	6	4.00	0.44	No
3.15	Conduct training for workers on sealant	3.14	3	5	7	5.00	0.44	No
4.1	Conduct structural and safety inspections for	3.15	5	8	12	8.33	1.00	Yes
4.2	Review quality of materials	4.1	3	5	8	5.17	0.69	No
4.3	Conduct system testing for	4.1	6	8	12	8.33	1.00	Yes
4.4	Perform final adjustments as needed based on	4.2, 4.3	3	5	8	5.17	0.69	No
4.5	Document results of quality inspections	4.4	2	3	4	3.00	0.11	No
4.6	Perform a comprehensi ve maintenance inspection of	4.3	5	7	10	7.33	0.69	Yes
4.7	Schedule additional repairs or adjustments	4.6	3	4	6	4.00	0.25	No
4.8	Finalize safety protocols for completed	4.5	2	3	5	3.00	0.25	No
5.1	Update equipment records and document condition for handover	4.7	2	3	4	3.00	0.11	Yes
5.2	Develop a punch list of items needing	5.1	3	5	7	5.00	0.44	No
5.3	Conduct follow-up inspections to ensure all adjustments	5.2	4	6	8	6.00	0.44	Yes
5.4	Coordinate final approvala	5.3	3	4	6	4.00	0.25	Yes
5.5	Perform mock drills for emergency	5.4	3	5	7	5.00	0.44	No
5.6	Organize a project review meeting for quality	5.5	2	3	4	3.00	0.11	No
5.7	Finalize and distribute quality	5.6	2	3	5	3.00	0.25	No
5.8	Develop a preventive maintenance schedule	5.4	3	5	8	5.17	0.69	Yes
5.9	Assign maintenance personnel for routine tasks	5.8	2	4	6	4.00	0.44	No
5.10	Perform monthly preventive maintenance	5.9	4	6	9	6.17	0.69	No
5.11	Conduct quarterly inspections	5.10	6	8	12	8.33	1.00	Yes
5.12	Schedule emergency repair	5.11	3	5	7	5.00	0.44	No
5.13	Review and update equipment	5.12	2	3	4	3.00	0.11	No
5.14	Establish spare parts inventory	5.13	3	4	6	4.00	0.25	No
5.15	Train maintenance staff on critical equipment	5.14	4	6	8	6.00	0.44	No
5.16	Conduct final equipment inspections	5.15	5	7	10	7.33	0.69	Yes
5.17	Archive all maintenance and project documentati	5.16	2	3	4	3.00	0.11	Yes

Gantt chart With Critical Path



► Red bars show tasks that impact the project's end date; delays here extend the overall timeline.

► Project runs from NOV 2024 to July 2025, covering planning, excavation, and final inspections.

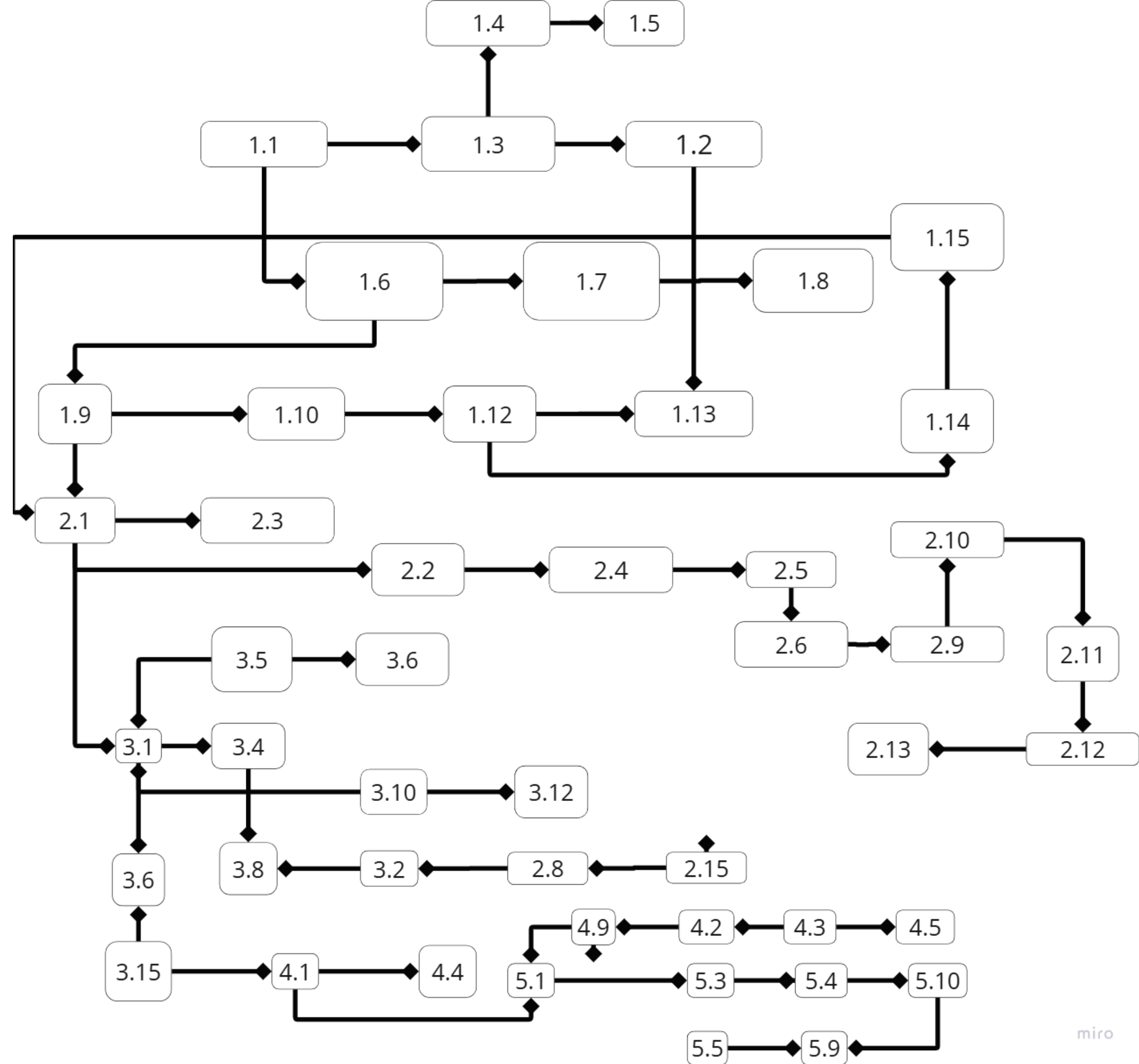
► Blue bars represent dependent tasks; focus remains on critical tasks to prevent delays.

Network Diagram

Sequential Task Flow and Dependencies:

The network diagram provides a visual representation of tasks across all project phases, showing how each task depends on others.

This ensures that key activities are completed in sequence, optimizing the project's workflow and preventing bottlenecks.



PERT Analysis for Task Durations



Optimistic Time	309 days
Pessimistic Time	610 days
Tm	410 days
Te	432 days
SD (σ)	50.17
Z value	0.94
P (Z=0.4207)	82.6%



To estimate the duration of each task
PERT formula: $(O + (4 \times M) + P) \div 0$.

Findings

Using the formula for 90% likelihood:
 $T_{90\%} = E(T) + (Z \times \sigma)$

Where:

$E(T)$ = Expected Time = 432 days
 Z = Z-value for 90% confidence = 1.28
 σ = Standard Deviation = 50.17 days

Substituting the values:

$T_{90\%} = 432 + (1.28 \times 50.17)$
 $= 432 + 64.13$
 $= 496.13$ days
 $T_{90\%} = 432 + (1.28 \times 50.17)$
 $= 432 + 64.13 = 496.13$ days

So, the project will be completed in 496.13 days with a 90% likelihood.

Metric	Value
Project Duration (Mean)	432 days.
Standard Deviation	50.17 days.
90% Likelihood	496.13days
15% Early Completion Time	9.85%.
10% Crash Completion Time	452.93 days

Completing the Project 15% Earlier

$T_{15\% \text{ early}} = E(T) \times (1 - 0.15) = 432 \times 0.85 = 367.2 \text{ days}$

$T_{15\% \text{ early}} = E(T) \times (1 - 0.15) = 432 \times 0.85 = 367.2 \text{ days}$

Now, calculate the Z-score for 367.2 days using the formula:

$$Z = \frac{T - E(T)}{\sigma} = \frac{367.2 - 432}{50.17} = -1.29$$

Substitute the values:

$$Z = \frac{367.2 - 432}{50.17} = -1.29$$

Looking up a Z-score of -1.29 in the Z-table, we get a 0.0985 or 9.85%.
So, the probability of completing the project 15% earlier is 9.85%.

$T_{\text{crash}} = E(T) \times (1 - 0.10) = 432 \times 0.90 = 388.8 \text{ days}$
 $T_{\text{crash}} = E(T) \times (1 - 0.10) = 432 \times 0.90 = 388.8 \text{ days}$

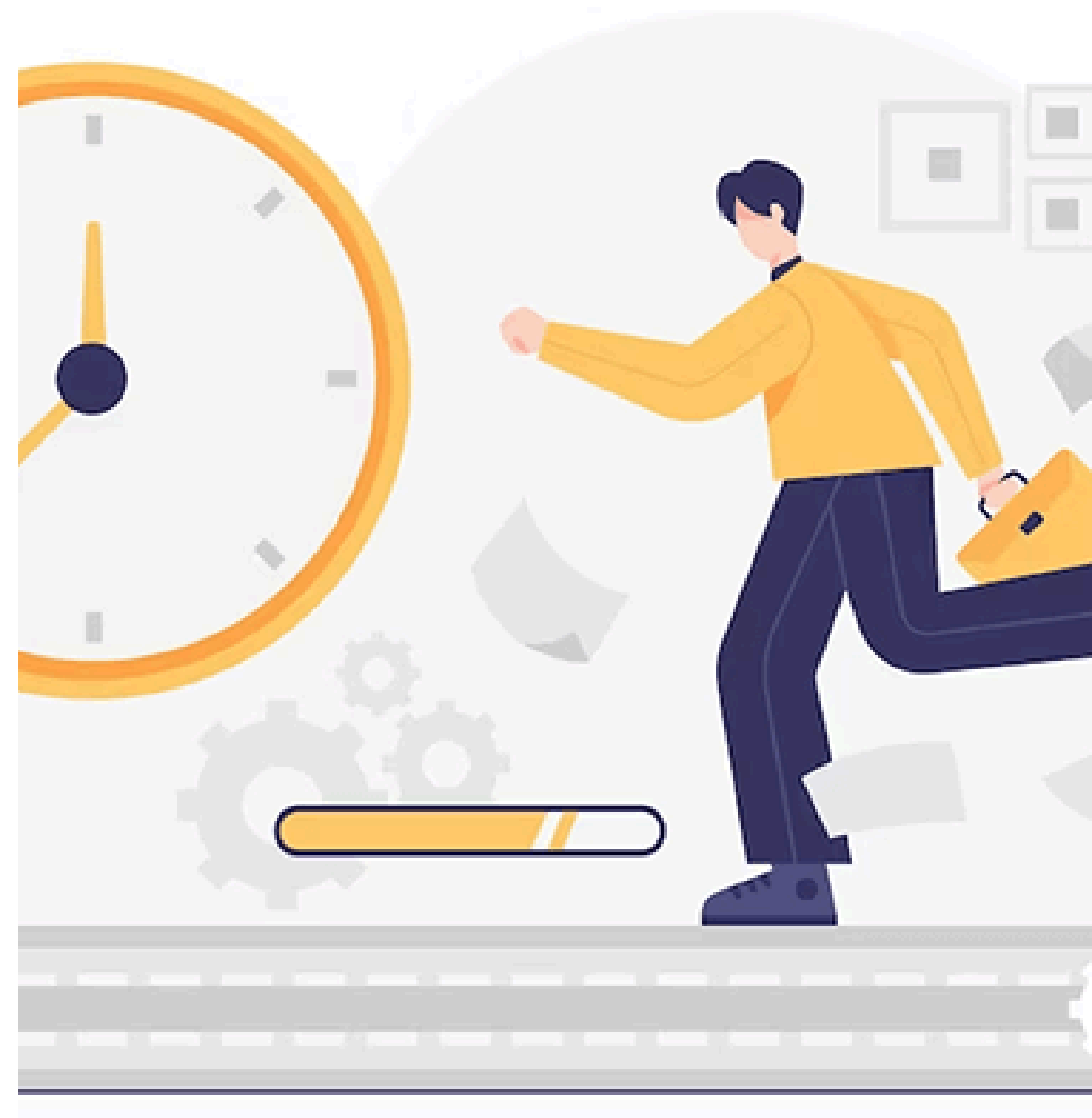
Crashing the Project by 10%

90% likelihood for the crashed duration:

$T_{90\% \text{ crash}} = 388.8 + (1.28 \times 50.17) = 388.8 + 64.13 = 452.93 \text{ days}$

$T_{90\% \text{ crash}} = 388.8 + (1.28 \times 50.17) = 388.8 + 64.13 = 452.93 \text{ days}$

Thus, after crashing the project by 10%, the project will be completed in 452.93 days with a 90% likelihood.



Monte Carlo Analysis

Monte Carlo Simulation Results

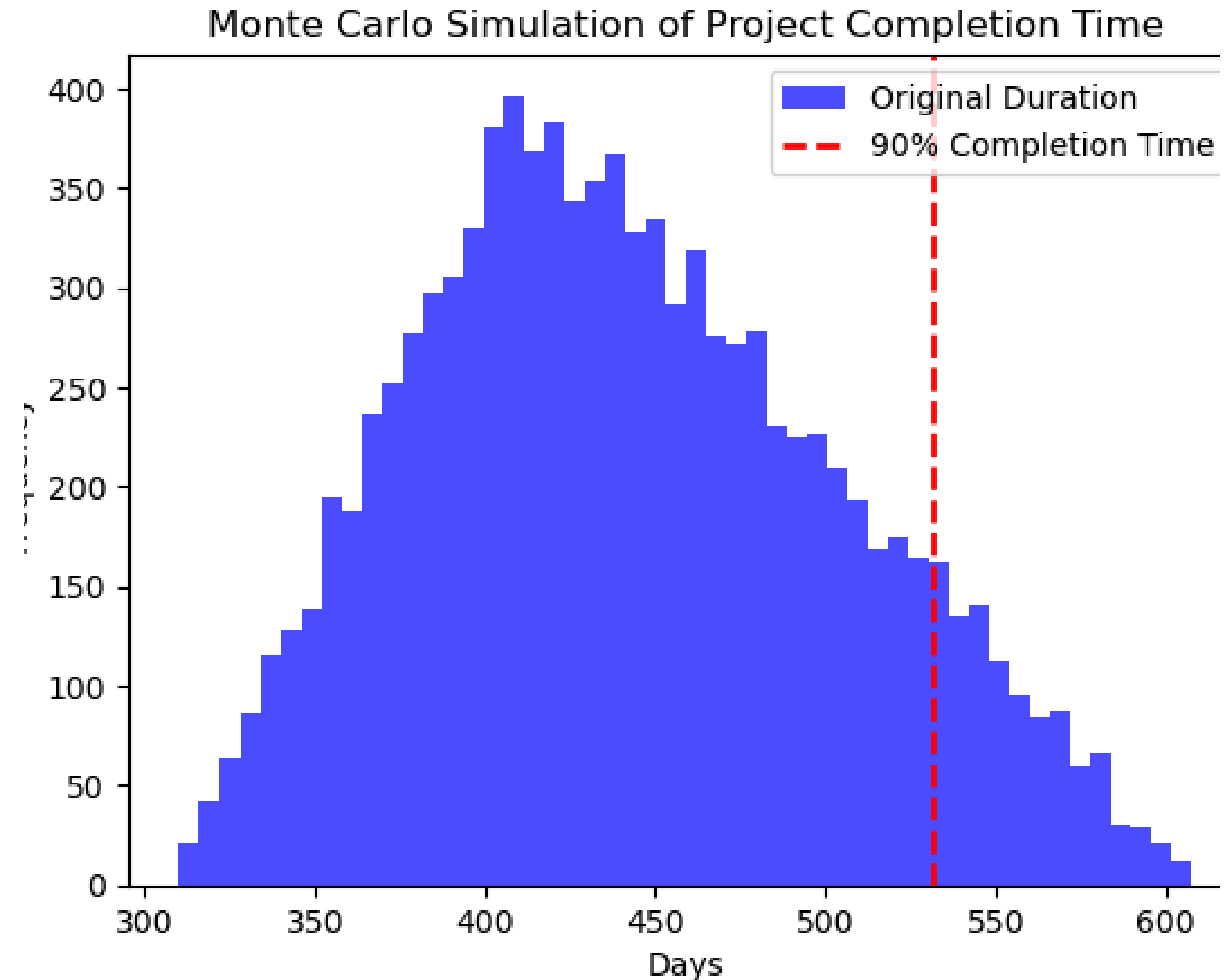
90% Likelihood Completion Time:
496.2 days

Based on the simulations, there's a 90% chance the project will finish within 496.2 days.

15% Early Completion Probability: 9.85%
There is a low chance (9.85%) that the project will finish 15% earlier than expected.

90% Likelihood After Crashing (10%):
453 days

After reducing the project duration by 10%, the project completion time with 90% confidence is now 453 days.

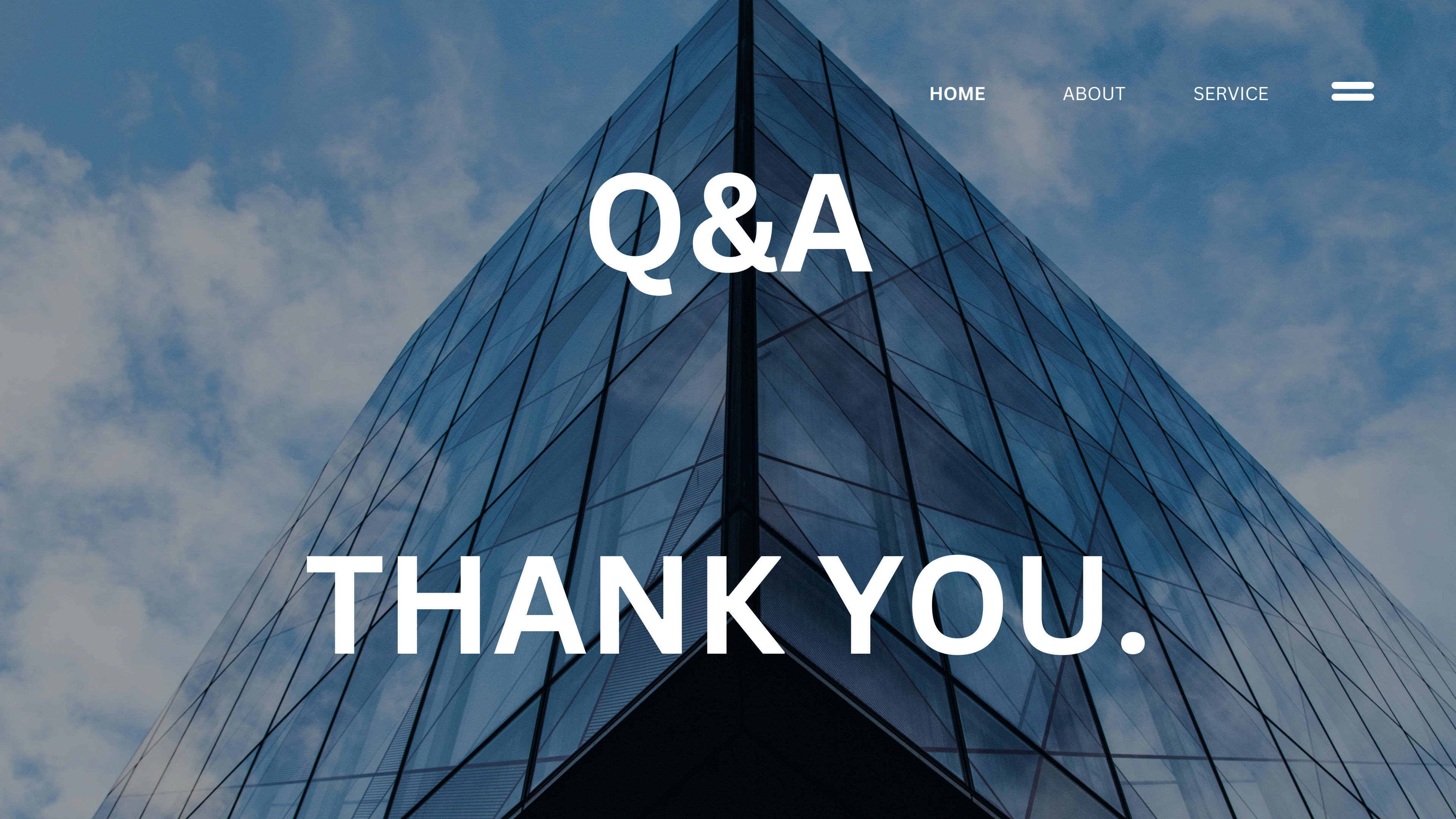


Risk	Risk description	Phase	Likelihood	Impact	Severity	Mitigation strategy	Risk owner
R1	Delays in material procurement	Planning	Medium	High	High	Identify multiple suppliers, pre-order critical items	Procurement Team
R2	Insufficient skilled labor	Planning	Medium	Medium	Medium	Conduct early recruitment, offer training	HR and Training Team
R3	Unexpected environmental issues	Excavation	High	High	High	Conduct thorough environmental assessments	Project Manager
R4	Equipment breakdown (Tunnel Boring Machine)	Excavation	Medium	High	High	Regular maintenance, have spare parts available	Maintenance Team
R5	Health and safety incidents	Excavation	Medium	High	High	Conduct regular safety briefings, provide PPE	Safety Officer
R6	Weather disruptions affecting concrete setting	Tunnel Lining	Medium	Medium	Medium	Schedule concrete work during stable weather periods	Site Manager
R7	Quality control failures in waterproofing	Tunnel Lining	Low	High	Medium	Perform frequent inspections and tests	Quality Assurance Team
R8	Failure to pass structural inspections	Quality Checks	Low	High	Medium	Schedule pre-inspections, correct issues proactively	Quality Control Team
R9	Scheduling conflicts with other construction teams	Maintenance	Medium	Medium	Medium	Use scheduling software, coordinate with other teams	Project Scheduler
R10	Lack of spare parts for maintenance	Maintenance	Medium	High	High	Set up a reorder system, maintain spare inventory	Inventory Management Team
R11	Cost overruns due to unexpected expenses	All Phases	Medium	High	High	Implement strict budget control, keep contingency fund	Finance Manager
R12	Delayed regulatory approvals	Planning	Low	High	Medium	Submit early applications, maintain close contact with regulators	Legal Team
R13	Miscommunication between teams	All Phases	Medium	Medium	Medium	Weekly progress meetings, use centralized communication tool	Project Coordinator
R14	Security breaches or vandalism on-site	Excavation	Low	High	Medium	Install security cameras, hire security personnel	Security Team
R15	Supplier quality issues with materials	Planning	Medium	Medium	Medium	Conduct supplier audits, establish quality standards	Procurement Team
R16	Delays in utility connections	Tunnel Lining	Low	Medium	Low	Coordinate early with utility companies	Site Manager
R17	Failure of waterproofing materials	Tunnel Lining	Low	High	Medium	Test materials beforehand, have backup materials	Quality Assurance Team
R18	Protests or public opposition	All Phases	Low	High	Medium	Engage with community, provide transparent updates	Public Relations Team
R19	Data loss in project management software	All Phases	Low	Medium	Low	Regular data backups, ensure access control	IT Support
R20	Failure to meet environmental standards	Quality Checks	Medium	High	High	Conduct periodic environmental assessments	Environmental Officer

Conclusion

1. Scope and plan of a sub project under Trans under Atlantic Tunnel.
2. Produced a work breakdown structure encompassing 69 activities.
3. Developed network diagram for the WBS and formulated tables showing activity, logic (ie precedence), and optimistic time, pessimistic time, most likely time' expected time, variance, slack etc.
4. The project will start on the 1st November 2024.
5. PERT plan with a Gantt Chart.
6. Calculated following calculations :
 - When the project will be available with 90% likelihood.
 - Probability of when the project will be available if it was to be 15% early.
 - Crashing project by 10% earlier.





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Q&A

THANK YOU.

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